Kubernetes Task

Task Description:

Setup minikube at your local and explore creating namespaces (Go through official documentation).

Setting up Minikube:

Installing docker, kubectl and minikube:

As I already have all the required tools, these are the version of docker, kubectl and minikube:

```
GHOST@GHOST: ~
GHOST@GHOST:~$ docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1~24.04.1
GHOST@GHOST:~$ kubectl version --client
Client Version: v1.35.2
Kustomize Version: v5.7.1
GHOST@GHOST:~$ minikube version
minikube version: v1.38.1
commit: c93a4cb9311efc66b90d33ea03f75f2c4120e9b0
GHOST@GHOST:~$
```

Starting docker driver using minikube:

```
GHOST@GHOST: ~
GHOST@GHOST:~$ minikube start --driver=docker
😄 minikube v1.38.1 on Ubuntu 24.04 (kvm/amd64)
✨ Using the docker driver based on user configuration
❗ Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.

❗ The requested memory allocation of 2842MiB does not leave room for system overhead (total system memory: 2842MiB). You may face stability issues.
💡 Suggestion: Start minikube with less memory allocated: 'minikube start --memory=2842mb'

📌 Using Docker driver with root privileges
👍 Starting "minikube" primary control-plane node in "minikube" cluster
🚜 Pulling base image v0.0.50 ...
💾 Downloading Kubernetes v1.35.1 preload ...
    > preloaded-images-k8s-v18-v1...:  272.45 MiB / 272.45 MiB  100.00% 15.64 M
    > gcr.io/k8s-minikube/kicbase...:  519.58 MiB / 519.58 MiB  100.00% 18.02 M

🔥 Creating docker container (CPUs=2, Memory=2842MB) ...
🐳 Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔗 Configuring bridge CNI (Container Networking Interface) ...
🔎 Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: storage-provisioner, default-storageclass
🏄 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
GHOST@GHOST:~$
GHOST@GHOST:~$
GHOST@GHOST:~$
```

Checking the docker container which we started, then docker image then the cluster info and node.

Listing the default namespaces and **creating a new namespace named dev.**

```
GHOST@GHOST: ~
GHOST@GHOST:~$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED         STATUS         PORTS
                                                                                                   NAMES
973d7f8d2c4f   gcr.io/k8s-minikube/kicbase:v0.0.50   "/usr/local/bin/entr…"   7 minutes ago   Up 7 minutes   127.0.0.1:32768->22/tcp, 127.0.0.1:32769->2376/
tcp, 127.0.0.1:32770->5000/tcp, 127.0.0.1:32771->8443/tcp, 127.0.0.1:32772->32443/tcp   minikube
GHOST@GHOST:~$ docker images
REPOSITORY                TAG       IMAGE ID       CREATED        SIZE
gcr.io/k8s-minikube/kicbase   v0.0.50   6da180ef5035   2 weeks ago    1.37GB
hello-world               latest    1b44b5a3e06a   7 months ago   10.1kB
GHOST@GHOST:~$ kubectl cluster-info
Kubernetes control plane is running at https://127.0.0.1:32771
CoreDNS is running at https://127.0.0.1:32771/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
GHOST@GHOST:~$ kubectl get nodes\
> ^C
GHOST@GHOST:~$ kubectl get nodes
NAME       STATUS   ROLES           AGE     VERSION
minikube   Ready    control-plane   7m32s   v1.35.1
GHOST@GHOST:~$
GHOST@GHOST:~$
GHOST@GHOST:~$ kubectl get namespaces
NAME              STATUS   AGE
default           Active   7m45s
kube-node-lease   Active   7m45s
kube-public       Active   7m45s
kube-system       Active   7m45s
GHOST@GHOST:~$ kubectl create namespaces dev
error: Unexpected args: [namespaces dev]
See 'kubectl create -h' for help and examples
GHOST@GHOST:~$ kubectl create namespace dev
namespace/dev created
GHOST@GHOST:~$
GHOST@GHOST:~$ kubectl get namespaces
NAME              STATUS   AGE
default           Active   8m9s
dev               Active   8s
kube-node-lease   Active   8m9s
kube-public       Active   8m9s
kube-system       Active   8m9s
GHOST@GHOST:~$
```

Creating, listing and deleting namespace:

```
GHOST@GHOST: ~
GHOST@GHOST:~$ kubectl create namespace dev
namespace/dev created
GHOST@GHOST:~$ kubectl get ns
NAME              STATUS   AGE
default           Active   11m
dev               Active   6s
kube-node-lease   Active   11m
kube-public       Active   11m
kube-system       Active   11m
GHOST@GHOST:~$
GHOST@GHOST:~$
GHOST@GHOST:~$ kubectl delete namespace dev
namespace "dev" deleted
GHOST@GHOST:~$
GHOST@GHOST:~$ kubectl get ns
NAME              STATUS   AGE
default           Active   11m
kube-node-lease   Active   11m
kube-public       Active   11m
kube-system       Active   11m
GHOST@GHOST:~$
GHOST@GHOST:~$
GHOST@GHOST:~$
```